Menus, Dialogs and UI Interaction

1. Develop an Android application implementing AppBar (Toolbar) and Options Menu.

Project Name: ToolbarMenuDemo

FileName: activity_main.xml

<?xml version="1.0" encoding="utf-8"?>

```
<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"

   android:id="@+id/main"

   android:layout_width="match_parent"

   android:layout_height="match_parent" android:orientation="vertical">

   <!-- Toolbar -->

   <androidx.appcompat.widget.Toolbar

       android:id="@+id/toolbar"

       android:layout_width="match_parent"

       android:layout_height="?attr/actionBarSize"

       android:background="?attr/colorPrimary"

       app:title="Toolbar Demo"

       app:titleTextColor="@android:color/white"/>

   <!--Main Content -->

   <TextView

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="AppBar and Options Menu Example"
```

```
        android:textSize="22sp"

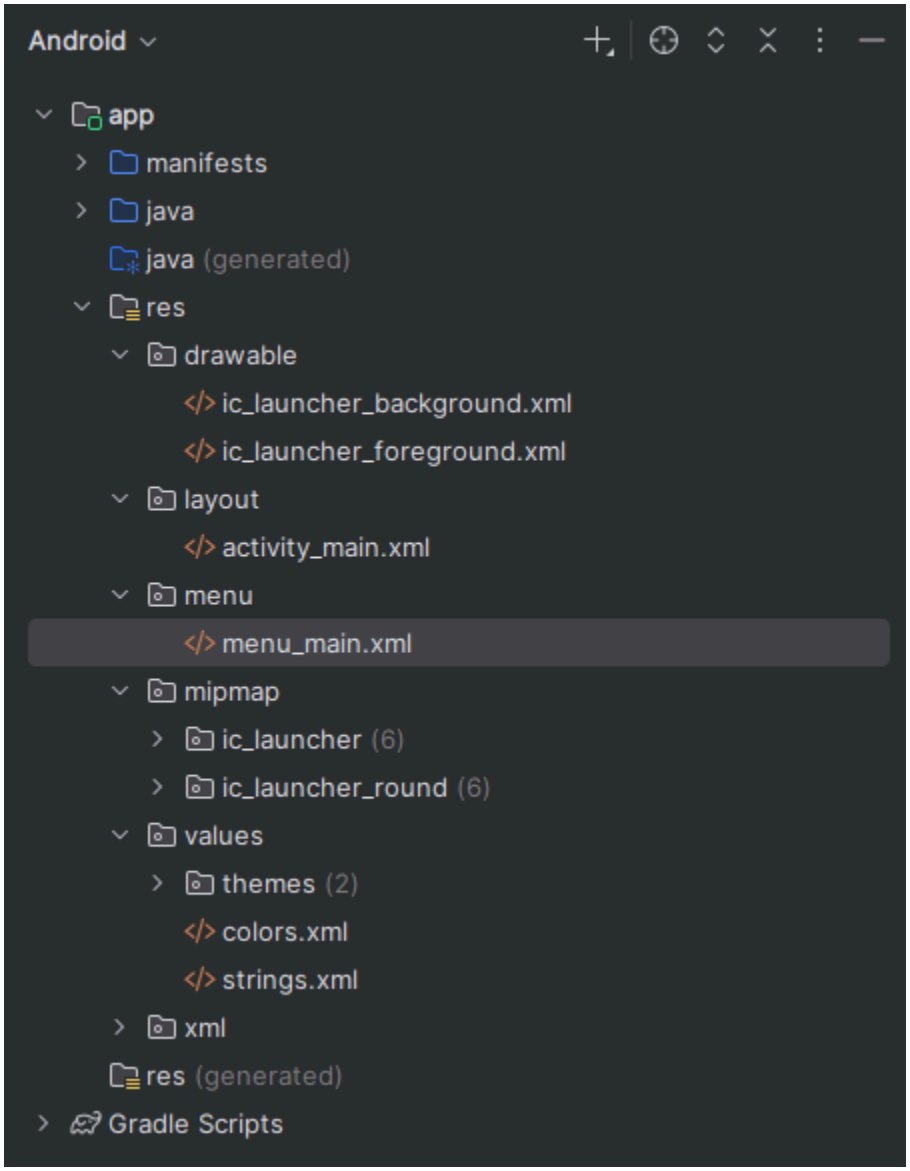
        android:padding="20dp"/>

</LinearLayout>
```

Create a Menu Resource File.
Right Click: Res → New→ Android Resource Directory

Directory Name: menu
Resource Type: menu

Create Menu XML File.
Right Click: menu New→ Menu Resource File Give name as **menu_main.xml**



File Name : menu_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<menu xmlns:android="http://schemas.android.com/apk/res/android"

   xmlns:app="http://schemas.android.com/apk/res-auto">



   <item

       android:id="@+id/menu_home"

       android:title="Home"

       app:showAsAction="never"/>



   <item

       android:id="@+id/menu_settings"

       android:title="Settings"

       app:showAsAction="never"/>

   <item

       android:id="@+id/menu_about"

       android:title="About"

       app:showAsAction="never"/>

   <item

       android:id="@+id/menu_logout"

       android:title="Logout"

       app:showAsAction="never"/>

</menu>
```

File Name: MainActivity.java.

```
package com.example.toolbarmenudemo;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;

public class MainActivity extends AppCompatActivity {

   Toolbar toolbar;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);
       toolbar = findViewById(R.id.toolbar);
       // Set Toolbar as AppBar
       setSupportActionBar(toolbar);
   }
   // Create Options Menu
   @Override
   public boolean onCreateOptionsMenu(Menu menu) {
```

```
        getMenuInflater().inflate(R.menu.menu_main, menu);

        return true;

    }

    // Handle Menu Item Click

    @Override

    public boolean onOptionsItemSelected(@NonNull MenuItem item) {

        int id = item.getItemId();

        if (id == R.id.menu_home) {

            Toast.makeText(this, "Home Selected", Toast.LENGTH_SHORT).show();

        } else if (id == R.id.menu_settings) {

            Toast.makeText(this, "Settings Selected",
Toast.LENGTH_SHORT).show();

        } else if (id == R.id.menu_about) {

            Toast.makeText(this, "About Selected", Toast.LENGTH_SHORT).show();

        } else if (id == R.id.menu_logout) {

            Toast.makeText(this, "Logout Selected", Toast.LENGTH_SHORT).show();

        }

        return true;

    }

}
```